

Unit 3 Homework — Equations & Inequalities

Complete independently · One/Two-Step · Distributive · Multi-Step · Special Cases · Inequalities · Word Problems

A. One- & Two-Step Equations

Problem 1

Solve: $x - 6 = 11$

17

Problem 2

Solve: $5x = 45$

9

Problem 3

Solve: $2x - 7 = 9$

8

Problem 4

Solve: $x/4 + 5 = 8$

12

Problem 5

Solve: $-3x + 2 = -10$

4

B. Distributive Property & Variables on Both Sides

Problem 1

Solve: $4(x - 2) = 20$

7

Problem 2

Solve: $2(3x + 1) = 26$

4

Problem 3

Solve: $6x + 5 = 3x + 20$

5

Problem 4Solve: $8x - 1 = 5x + 17$ 6**C. Multi-Step Equations****Problem 1**Solve: $2(x + 5) = 3x - 1$ 11**Problem 2**Solve: $6x + 2 = 3(x + 8)$ 22/3**Problem 3**Solve: $10 - x = 2x + 1$ 3**D. Special Cases — No Solution or Infinite Solutions****Problem 1**Solve: $5x - 2 = 5x - 2$ — how many solutions?infinite**Problem 2**Solve: $3x + 7 = 3x - 4$ — how many solutions?no solution**Problem 3**Solve: $2(x + 6) = 2x + 12$ — how many solutions?infinite**E. Linear Inequalities****Problem 1**Solve: $x - 5 < 2$ $x < 7$

Problem 2

Solve: $4x \geq 32$

$$\underline{x \geq 8}$$

Problem 3

Solve: $-5x < 20$ (careful!)

$$\underline{x > -4}$$

Problem 4

Solve: $2x + 7 \leq 19$

$$\underline{x \leq 6}$$

Problem 5

Solve: $-4x - 3 > 9$ (careful!)

$$\underline{x < -3}$$

F. Word Problems**Problem 1**

A phone plan costs \$20 plus \$0.05 per text. If the bill was \$35, how many texts were sent?

$$\underline{300}$$

Problem 2

You have \$60 and want to buy shirts that cost \$12 each. What's the max number of shirts you can buy?

$$\underline{5}$$